BEFORE · Approved reference · Naor_Larian_CV(5).pdf

Naor Larian

FPGA & RTL | ELECTRICAL ENGINEERING STUDENT

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Central Israel
linkedin.com/in/naor-larian

+ SUMMARY

Final-year Electrical & Electronics Engineering student (expected 2027) with hands-on VHDL and Vivado experience. Designed and implemented a working FPGA system on Artix-7 hardware, integrating ESP32 control over UART with real-time VGA graphics. Nearly four years of technical experience in live satellite communication systems.

+ FPGA PROJECTS

2026 / Cyber Gladiators, Independent FPGA Project

- Designed a modular **VHDL game engine** with 640 × 480 VGA output, sprite rendering, FSM-based control, collision detection, and gameplay logic.
- Integrated an **ESP32 controller through UART** and demonstrated the complete design running on a physical Digilent Basys 3 (Artix-7) board.
- **Verified RTL modules through simulation**, synthesized and implemented the design in Xilinx Vivado, and debugged UART-to-VGA integration on physical FPGA hardware.

2026 / RGB Median Filter, Afeka Team Project

- Developed **internal VHDL components** and integrated FSM control, data buffers, and RAM/ROM handling for a 3 × 3 image median filter.
- Verified module and top-level functionality using RTL simulations and image test data.

+ EXPERIENCE

2022 - Present / AMOS-Spacecom, Technical Support Engineer

- Investigate and troubleshoot **15–25 technical incidents per week** across live satellite systems, correlating RF spectrum measurements, modem indicators, and link performance data.
- Support approximately **10–15 CPI tests per month**, performing operator-side RF measurements and guiding customers through testing procedures.
- Developed and refined **technical SOPs** to improve troubleshooting consistency and make procedures and critical information easier to find.

+ TECHNICAL SKILLS

FPGA & RTL: VHDL, FSM, RTL simulation, testbenches, Vivado, Artix-7 / Basys 3

Interfaces & systems: UART, ESP32, VGA, RAM/ROM, hardware integration

Programming & tools: C, Python, MATLAB/Simulink (academic); spectrum analyzer (professional)

+ EDUCATION

2027 (expected) / B.Sc. Electrical & Electronics Engineering

Afeka Tel Aviv Academic College of Engineering. Computers & Communications specializations.

2022 / Practical Engineer Diploma, Electrical Engineering

Handessaim Tel Aviv, graduated with honors.

+ RELEVANT STUDIES

Advanced Hardware Design, Digital Signal Processing, Communication Systems, Signals & Systems.

+ LANGUAGES

Hebrew / Native

English / Professional working proficiency

AFTER · System Integration v1 · CV_Design_Master_2026_11

Naor Larian

SYSTEM INTEGRATION & VALIDATION | ELECTRICAL ENGINEERING STUDENT

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Central Israel

+ SUMMARY

Final-year B.Sc. Electrical & Electronics Engineering student (expected 2027) with nearly four years of hands-on experience on **live satellite communication systems** at AMOS-Spacecom: fault isolation, RF spectrum measurements, link diagnostics, interference investigation and CPI testing, in close coordination with customers and engineering teams. Adds **FPGA hardware integration** experience from a VHDL/Artix-7 system demonstrated on physical hardware.

+ EXPERIENCE

Technical Support Engineer — Dec 2022 – Present
AMOS-Spacecom · Satellite Operator

- Troubleshoot **15–25 technical incidents per week** on live satellite communication systems, correlating RF spectrum measurements, modem indicators and link performance data to isolate faults.
- Investigate **RF interference** and link degradation using spectrum analysis and step-by-step fault isolation, coordinating resolution with customers and internal engineering teams.
- Support approximately **10–15 CPI tests per month**, performing operator-side RF measurements and guiding customers through test procedures during commissioning-related activities.
- Provide operational integration support for **VSAT systems**, coordinating between customers and engineering teams.
- Wrote and refined **technical SOPs** that made troubleshooting more consistent and critical procedures faster to locate.

+ ENGINEERING PROJECTS

Cyber Gladiators – Independent FPGA Project — 2026
VHDL · Xilinx Vivado · Basys 3 (Artix-7) · ESP32 · UART · VGA

- Designed a modular **VHDL system** with 640 × 480 VGA output, sprite rendering, FSM-based control, collision detection and game logic.
- **Integrated an ESP32 controller over UART** and debugged the UART-to-VGA data path on physical FPGA hardware.
- Verified RTL modules in simulation, synthesized and implemented the design in Vivado, and **demonstrated the full system on a Basys 3 board**.

RGB Median Filter – Afeka Team Project — 2026
VHDL · FSM · RAM/ROM · RTL Simulation · Testbenches

- Developed internal **VHDL components** and integrated FSM control, data buffers and RAM/ROM handling for a 3 × 3 RGB median filter.
- Verified module- and top-level behavior through **RTL simulation and testbenches** using image test data.

+ EDUCATION

Expected 2027
B.Sc. Electrical & Electronics Engineering
Afeka Tel Aviv Academic College of Engineering
Specializations: Computers & Communications

2022
Practical Engineer Diploma, Electrical Engineering
Handessaim Tel Aviv
Graduated with honors

+ TECHNICAL SKILLS

Systems & RF
RF spectrum measurements, satellite link diagnostics, VSAT systems, interference investigation, fault isolation, CPI testing

Integration & Verification
Hardware integration, on-board debugging, RTL simulation, testbenches, technical SOPs

FPGA & Digital
VHDL, FSM design, Xilinx Vivado, Artix-7 / Basys 3

Interfaces
UART, VGA, ESP32, RAM/ROM

Programming & Tools
Spectrum analyzer (professional); C, Python, MATLAB/Simulink (academic)

+ RELEVANT COURSEWORK

Communication Systems, Digital Signal Processing, Signals & Systems, Advanced Hardware Design

+ LANGUAGES

Hebrew – Native
English – Professional working proficiency